

Taller: Control Robusto y Estocástico

Tema: Incertidumbre No Estructurada

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EXERCISE

G_0 = nominal model used to design the controller

G_1 to G_6 = possible physical plants or experiments

W = Weighting function covering discrepancies

Δ = unknown dynamics

```
% Part A: Nominal model

clear, clc, close all;

s = tf('s');

G0 = 1/(0.2*s + 1); %Nominal Servomotor Model
G0
```

```
pole(G0)
pzmap(G0)
grid on
```

POSSIBLE UNCERTAIN BEHAVIOR

In a real-world application, G_1 , G_2 , ..., G_6 could be:

- six experiments on the same motor,
- six manufactured motors,
- the motor under different loads,
- measurements at different temperatures,
- different operating points.

```
% Part B: Creation of uncertain plants
```

```

dyn1 = 80/(s + 80);
G1 = G0*dyn1;
G1

wn2 = 60;
zeta2 = 0.18;
dyn2 = wn2^2/(s^2 + 2*zeta2*wn2*s + wn2^2);
G2 = G0*dyn2;
G2

Tdelay3 = 0.006;
[numD3,denD3] = pade(Tdelay3,2);
dyn3 = tf(numD3,denD3);
G3 = G0*dyn3;
G3

dyn4 = 100/(s + 100);
wn4 = 90;
zeta4 = 0.25;
dyn41 = wn4^2/(s^2 + 2*zeta4*wn4*s + wn4^2);
G4 = G0*dyn4*dyn41;
G4

Tdelay5 = 0.004;
[numD5,denD5] = pade(Tdelay5,2);
dyn5 = tf(numD5,denD5);
dyn51 = 70/(s + 70);
G5 = 1.03*G0*dyn5*dyn51;
G5

wn6 = 110;
zeta6 = 0.15;
dyn6 = wn6^2/(s^2 + 2*zeta6*wn6*s + wn6^2);
G6 = 0.97*G0*dyn6;
G6

Garray = stack(1,G1,G2,G3,G4,G5,G6);
Garray

```

Although we use different equations to create these plants, it is assumed that the designer does not know them. The designer only has the results of six tests.

% Part C: Step response

```

t = 0:0.01:5;

figure
step(G0,Garray,t)
grid on

title('Nominal model and possible plants');
ylabel('Velocity');

```

From Class Slides:

For multiplicative uncertainty we take the relative error

$$E_i(j\omega) = \frac{G_i(j\omega) - G_0(j\omega)}{G_0(j\omega)}$$

% Part D: Frequency response

```

w = logspace(-1,3,300);
Gdata = frd(Garray,w);

Err_i = (Gdata - G0)/G0;

figure
bodemag(Err_i,{0.1,1000})
grid on

title('Relative Multiplicative Errors');
xlabel('Frequency');
ylabel('Magnitude');

```

In low frequencies, the magnitude of the relative error is less than -20dB. We can state that

$$G_i(j\omega) \approx G_0(j\omega)$$

because

$$|E_i(j\omega)| \approx 0.$$

But when ω approaches 60, 80, 90, or 110 rad/s, the neglected dynamics no longer appear equal to one. So

$$G_i(j\omega) \not\approx G_0(j\omega)$$

and the relative error increases.

We now search for a function $W(s)$ whose magnitude covers all errors.

$$|E_i(j\omega)| \leq |W(j\omega)|$$

for all the plants and frequencies in the problems.

```
% Part E: Fit the W.  
  
orderW = 3;  
[Gunc,Info] = uncover(Gdata,G0,orderW,'InputMult');  
W = Info.W1;  
tf(W)  
zpk(W)
```

The obtain system has the form

$$G_{unc}(s) = G_0(s)[1 + W(s)\Delta(s)]$$

```
figure  
bodemag(Err_i, '--', W, {0.1, 1000});  
grid on  
  
title('Relative errors and Weighting function W');  
legend('Errors', 'Third order W Function', 'Location', 'southeast')
```

```
Gunc.NominalValue  
Gunc.Uncertainty
```

Try higher order

```
orderW = 5;  
[Gunc,Info] = uncover(Gdata,G0,orderW,'InputMult');  
W = Info.W1;  
  
figure  
bodemag(Err_i, '--', W, {0.1, 1000});  
grid on  
  
title('Relative errors and Weighting function W');  
legend('Errors', 'High order W Function', 'Location', 'southeast')
```

The object created by MATLAB not only contains the six original plants but also includes all plants of the form:

$$G(s) = G_0(s) [1 + W(s)\Delta(s)], \quad \|\Delta\|_\infty \leq 1.$$

This is what the unstructured uncertainty approach addresses. We view the original plant as a large family of plants with many added unknown dynamics, all of which are bounded by $W(s)$.

Therefore, for robust control, the problem is formulated as follows: "**designing a controller that works not only for the nominal model but also for all unknown dynamics consistent with the observed discrepancies.**"